# Continuous Gradient-Energy Importance Accumulation Reduces Catastrophic Forgetting: An Empirical Audit of EWC's Fisher Snapshot Assumption

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**Abstract**

Elastic Weight Consolidation (EWC) estimates per-parameter importance as a uniform average of squared gradients from the final training batches of each task—a Fisher snapshot that conflates high-noise early-training gradients with the convergence-phase curvature that Fisher information actually captures. We propose *continuous gradient-energy importance accumulation*: an exponential moving average (EMA) of squared gradients maintained throughout training, which we show converges to the diagonal Fisher information with exponentially decaying bias while down-weighting early-training noise. The resulting regularizer, TCL, applies an elastic penalty using EMA-derived importance weights in place of EWC's terminal snapshot.

On Split-CIFAR-10 (ResNet-18, 5 tasks, $n$=5 seeds, 40 epochs/task), TCL achieves mean catastrophic forgetting 0.116 (95% CI $[0.080, 0.153]$) versus SGD's 0.233 ($[0.229, 0.237]$), a reduction of 0.117 ($p$=0.0012, Cohen's $d$=3.68, 5/5 seeds, Bonferroni-corrected $k$=4). Against EWC ($\lambda$=100), TCL shows directional improvement $\Delta$=$-0.075$ ($p$=0.019, $d$=1.70, 5/5 seeds), which does not survive Bonferroni correction ($\alpha/k$=0.0125, $p$=0.019 > 0.0125).

A pre-registered mechanism ablation definitively establishes that the elastic penalty is the sole productive component: a governor-alone condition (regime-detection without penalty) performs *worse* than plain SGD (0.250 vs. 0.219), and the thermodynamic learning-rate adjustment never activates in any production trace ($\rho$=0, $\sigma$=0). We conclude that continuous importance accumulation, not thermodynamic regime detection, explains the empirical benefit.

## 1 Introduction

Neural networks trained sequentially on a stream of tasks suffer from *catastrophic forgetting*: optimizing for a new task overwrites parameters critical for previously learned tasks [8]. Regularization-based continual learning methods address this by penalizing changes to parameters deemed important for past tasks, with the importance estimated from the Fisher information matrix [4] or path integrals of parameter movements [12].

The dominant approach, EWC [4], approximates the diagonal Fisher by averaging squared gradients over the final $N$ training batches immediately after each task completes. This snapshot has two structural weaknesses. First, gradients early in training reflect random initialization and high loss rather than the task's true parameter importance; mixing these with late-training gradients biases the importance estimate upward for parameters with large early-training noise. Second, the snapshot is computed *once* from a terminal window and never updated as training continues on future tasks, discarding information from the entire non-terminal training trajectory.

**This paper.** We propose replacing EWC's terminal snapshot with a continuous EMA of squared gradients—the *ThermalImportance* accumulator. We show (Theorem 1) that this accumulator converges to the diagonal Fisher information with exponentially decaying bias, and that EWC's snapshot estimator is the limiting case $\beta \to 1$. Under non-stationary gradient processes (as occur in all neural network training), the EMA's recency weighting provides a strictly smaller non-stationarity bias than EWC's uniform average (Remark 1).

We evaluate the resulting regularizer, TCL,[1] against EWC, SI, and SGD on Split-CIFAR-10 using a pre-registered controlled experiment with $n$=5 random seeds and corrected statistical methodology (t-distribution confidence intervals, Bonferroni family-wise correction, non-central $t$ power analysis).

Our contributions are:

1. **Formal analysis.** We prove that the ThermalImportance EMA is an online, continuously-updated approximation to the diagonal Fisher information, with EWC as the $\beta \to 1$ limiting case and exponentially smaller non-stationarity bias (Section 3).

2. **Controlled empirical comparison.** On Split-CIFAR-10 with corrected statistical methodology, TCL reduces forgetting by 0.117 relative to SGD ($p$=0.0012, $d$=3.68, Bonferroni-significant at $k$=4). Relative to EWC, the improvement is directional ($p$=0.019, $d$=1.70) but does not survive family-wise correction (Section 4).

3. **Mechanism ablation.** A pre-registered four-condition ablation establishes that the elastic penalty is the sole productive mechanism: governor-alone performs worse than SGD, and the LR adjustment never fires in production (Section 4).

## 2 Related Work

**Catastrophic forgetting and evaluation scenarios.** Sequential learning in artificial neural networks induces catastrophic interference [8]: gradient-based updates that minimize a new task's loss overwrite weight configurations that supported prior tasks. Van de Ven and Tolias [11] identify three evaluation scenarios: task-incremental (task identity provided at test time), class-incremental (task identity withheld), and domain-incremental. This paper addresses the *task-incremental* scenario exclusively.

**Regularization-based methods.** EWC [4] adds an elastic term to the loss function that penalizes changes to parameters with high Fisher information: $\mathcal{L}_{\text{EWC}} = \mathcal{L}_{\text{CE}} + \frac{\lambda}{2}\sum_i \hat{F}_{ii}(\theta_i - \theta_i^*)^2$, where $\hat{F}_{ii}$ is the diagonal of the empirical Fisher estimated from a post-task snapshot. EWC's importance estimate is a uniform average of squared gradients over the final $N$ batches, discarding the full training trajectory.

Synaptic Intelligence (SI) [12] accumulates importance online via path integrals of parameter movements during training, providing a stronger prior over slow-moving parameters. However, SI at published default hyperparameters collapses to chance-level accuracy on our benchmark (Section 4), consistent with known sensitivity to the regularization strength $c$.

Memory Aware Synapses [1] estimates importance from the gradient of the output (rather than the loss), reducing dependency on label information. All three methods share the structural assumption that importance can be reliably estimated from a finite sample; our work argues that for EWC specifically, continuous accumulation is more principled.

**Replay-based methods.** Replay methods maintain a memory buffer of past examples [6, 9] or generate synthetic past data [10]. Dark Experience Replay (DER++) [2] stores soft logits alongside raw inputs to better preserve the model's decision boundary. The current work does not compare against replay methods; DER++ and LwF baselines are included in the pre-registered Phase 2 replication plan but have not yet been run.

**Fisher information approximations.** The diagonal Fisher information $F_{ii} = \mathbb{E}_x[(\partial\mathcal{L}/\partial\theta_i)^2]$ is the cornerstone of EWC's importance weighting. Efficient diagonal approximations and Kronecker-factored estimates have been studied for second-order optimization [7], but continual learning has relied primarily on diagonal estimates computed from task-end snapshots. Our contribution is to show that replacing the snapshot with a continuously maintained EMA yields a formally better approximation while remaining computationally identical in per-parameter cost.

[1] TCL originally stood for *Thermodynamic Continual Learning*, reflecting a physical intuition that high-gradient parameters are "hot" and should be stabilized. Mechanism ablation (Section 4) shows the thermodynamic regime-detection component does not contribute to performance. We retain the acronym but clarify that the operative mechanism is purely gradient-energy elastic regularization.

# 3 Method

## 3.1 Background: EWC's Fisher Snapshot

Let $\theta \in \mathbb{R}^d$ denote network parameters after training on a task, and let $\mathcal{D}_{\text{task}}$ be the task data distribution. The diagonal Fisher information for parameter $i$ is $F_{ii} = \mathbb{E}_{x\sim\mathcal{D}_{\text{task}}}[g_i^2]$, where $g_i = \partial\mathcal{L}(\theta; x)/\partial\theta_i$.

EWC approximates $F_{ii}$ by the empirical mean over the final $N$ training batches:

$$\hat{F}_{ii}^{\text{EWC}} = \frac{1}{N}\sum_{t=T-N+1}^{T} g_i(t)^2. \tag{1}$$

This estimate is unbiased only if gradients are stationary throughout training. In practice, gradients decay substantially as training converges: early-training gradients reflect large loss curvature at random initialization, while late-training gradients reflect the convergence-phase curvature that Fisher information captures. A uniform window containing both phases overestimates importance for parameters with large early-training noise.

## 3.2 ThermalImportance EMA Accumulator

TCL maintains a running EMA of squared gradients throughout training. For parameter $i$ at training step $T$:

$$v_i^{(T)} = (1-\beta)\sum_{t=1}^{T} \beta^{T-t}\, g_i(t)^2, \tag{2}$$

where $\beta \in (0,1)$ is the EMA decay constant. This accumulator is updated continuously as training proceeds, requiring no designated importance-extraction phase after task completion.

**Theorem 1** (EMA convergence to diagonal Fisher). *Let $g_i(t) \overset{\text{i.i.d.}}{\sim} \mathcal{D}_{g_i}$ with $\mathbb{E}[g_i(t)^2] = F_{ii} < \infty$. For the EMA accumulator (2):*

$$\mathbb{E}[v_i^{(T)}] = F_{ii}(1-\beta^T). \tag{3}$$

*The bias $|\mathbb{E}[v_i^{(T)}] - F_{ii}| = F_{ii}\,\beta^T \to 0$ exponentially as $T \to \infty$.*

*Proof.* By linearity of expectation, $\mathbb{E}[v_i^{(T)}] = (1-\beta)\sum_{t=1}^{T}\beta^{T-t}F_{ii} = F_{ii}(1-\beta)\cdot\frac{1-\beta^T}{1-\beta} = F_{ii}(1-\beta^T)$. □

**Corollary 2** (EWC as limiting case). *As $\beta \to 1$ with $T$ fixed, the EMA weight $(1-\beta)\beta^{T-t}$ converges to the uniform weight $1/T$ for each $t$. Hence $v_i^{(T,\beta)} \to \frac{1}{T}\sum_{t=1}^{T} g_i(t)^2 = \hat{F}_{ii}^{\text{unif}}$, which is EWC's snapshot estimator applied over a full-training window.*

**Remark 1** (Non-stationarity advantage). *Under a gradient process that decays from initial magnitude $M$ to stationary magnitude $m \ll M$, EWC's uniform estimate on a window containing both phases has non-stationarity bias $O((M-m)/N)$. The EMA's exponential recency weighting yields bias $O(M\beta^T)$, which vanishes exponentially and is strictly smaller for large $T$.*

## 3.3 Elastic Penalty

After completing task $k$, TCL freezes $v_i^{(T_k)}$ as the importance weight for task $k$ and appends an elastic term to subsequent task losses:

$$\mathcal{L}_{\text{TCL}} = \mathcal{L}_{\text{CE}} + \lambda \sum_{k<\text{current}} \sum_{i=1}^{d} v_i^{(T_k)} \cdot \left(\theta_i - \theta_i^{(k*)}\right)^2, \tag{4}$$

where $\theta^{(k*)}$ is the parameter snapshot taken at the end of task $k$. The penalty coefficient $\lambda$ is a global hyperparameter shared across all tasks.

### 3.4 Governor Component (Non-Contributing)

TCL was originally designed with an additional *governor* component: a regime-detection mechanism that measures the ratio $\rho$ of current gradient magnitude to a reference magnitude $\sigma^*$, and adjusts the learning rate when $\rho$ falls below a threshold indicating convergence. Our mechanism ablation (Section 4) demonstrates that: (a) the governor-alone condition performs *worse* than plain SGD, and (b) $\rho = 0$, $\sigma = 0$ in every production trace examined, indicating the learning-rate adjustment has never fired. Equation (4) represents the operative TCL algorithm; the governor is a dead branch.

### 3.5 Connection to Fisher-EMA Theory

Equations (1) and (2) frame the TCL/EWC comparison as a statistical estimation problem: both methods estimate $F_{ii}$ from the same sequence of squared gradients, but with different weighting schemes. EWC uses a uniform average over a terminal window; TCL uses an exponentially weighted average over the full training history. Corollary 2 establishes EWC as a special case, and Theorem 1 and Remark 1 provide the theoretical motivation for preferring the EMA under non-stationary gradients.

## 4 Experiments

### 4.1 Experimental Setup

**Dataset and protocol.** We evaluate on Split-CIFAR-10 [5] in the task-incremental scenario: CIFAR-10 is partitioned into 5 tasks of 2 classes each, with task identity provided at test time. We use a ResNet-18 [3] backbone with a separate classification head per task. All experiments use 40 training epochs per task, batch size 32, and seeds $\{42, 0, 1, 2, 3\}$ ($n$=5 per condition). This protocol is identical for all methods, removing architecture and training-duration confounds.

**Metrics.** We report two primary metrics: *Catastrophic forgetting* ($_{\text{fgt}}$), defined as the average decrease in per-task accuracy after all tasks are trained: Forgetting $= \frac{1}{T-1}\sum_{k=1}^{T-1}(A_{k,k} - A_{T,k})$, where $A_{i,j}$ is accuracy on task $j$ after training task $i$. We also report final *accuracy* ($_{\text{acc}}$), the average accuracy across all tasks after training completes, to detect degenerate low-forgetting solutions that achieve low forgetting through accuracy collapse.

**Statistical methodology.** All confidence intervals use the $t$-distribution with $n-1$=4 degrees of freedom ($t_{4,0.975} = 2.776$), replacing the common but incorrect $z = 1.96$ approximation that assumes $n \to \infty$. Pairwise comparisons use paired one-tailed $t$-tests with Bonferroni family-wise correction: for $k$=4 simultaneous comparisons within Phase 10 (TCL vs. EWC, SI, SGD, and an SI accuracy-collapse diagnostic), the corrected significance threshold is $\alpha/k = 0.05/4 = 0.0125$. Power is computed via the non-central $t$-distribution (ncp $= -d\sqrt{n}$ for one-tailed tests).

**Baselines.** We compare against: **EWC** with $\lambda$=100 (the published default [4]), **SI** with default hyperparameters [12], and plain **SGD** (no regularization) as a lower bound. DER++ and LwF are included in the pre-registered Phase 2 replication plan but their GPU experiments have not yet run; we omit them from current results.

### 4.2 Phase 10: Main Comparison (Split-CIFAR-10)

Table 1 reports forgetting and accuracy for all four conditions across $n$=5 seeds. All confidence intervals use the $t$-distribution as described above.

**TCL vs. SGD.** TCL reduces forgetting by 0.117 relative to SGD ($p$=0.0012, Cohen's $d$=3.68, 5/5 seeds, 100% power at $n$=5). This result survives Bonferroni correction and constitutes the single confirmed headline finding of this paper.

Table 1: Split-CIFAR-10 main comparison (Phase 10). $n$=5 seeds, 95% CI via $t$-distribution ($t_{4,0.975}$=2.776). Bonferroni-corrected significance threshold $p < 0.0125$ ($k$=4). ⋆: Bonferroni-significant. †: directional ($p < 0.05$, uncorrected). Forgetting: lower is better. Accuracy: higher is better.

| | Forgetting ↓ | | Accuracy ↑ | |
|---|---|---|---|---|
| Method | Mean | 95% CI | Mean | 95% CI |
| TCL | 0.116 | [0.080, 0.153] | 0.760 | [0.729, 0.791] |
| EWC ($\lambda$=100) | 0.191 | [0.162, 0.220] | 0.730 | [0.702, 0.757] |
| SI (default) | 0.092 | [0.091, 0.093] | 0.500 | [0.500, 0.500] |
| SGD | 0.233 | [0.229, 0.237] | 0.697 | [0.693, 0.701] |

Table 2: Pairwise comparisons versus TCL (Phase 10, $n$=5, one-tailed paired $t$-test). $\Delta < 0$ means TCL has lower (better) forgetting.

| Comparison | $\Delta$ | $p$ (uncorr.) | Cohen's $d$ | Seeds TCL better |
|---|---|---|---|---|
| TCL vs. SGD | −0.117 | 0.0012⋆ | 3.68 | 5/5 |
| TCL vs. EWC | −0.075 | 0.019† | 1.70 | 5/5 |
| TCL vs. SI | +0.024 | 0.142 | 0.82 | 1/5 |

⋆ Survives Bonferroni ($p < 0.0125$, $k$=4). † Directional ($p < 0.05$ uncorrected); does *not* survive Bonferroni.

**TCL vs. EWC.** TCL reduces forgetting by 0.075 relative to EWC with default hyperparameters ($p$=0.019, $d$=1.70, 5/5 seeds, 92.5% power at $n$=5). This result does **not** survive Bonferroni correction ($p$=0.019 > 0.0125) and must be treated as directional evidence. At $d$=1.70 and $n$=5, the experiment had 92.5% power, meaning the failure to pass Bonferroni correction reflects the stringency of the family-wise threshold rather than a weak effect.

**SI accuracy collapse.** SI at published default hyperparameters achieves mean forgetting 0.092, *lower* than TCL's 0.116. However, SI's accuracy of 0.500 (chance level for a 2-class-per-task benchmark) is constant across all 5 seeds with zero variance, indicating catastrophic collapse rather than genuine retention. A method that predicts randomly on every task has trivially low forgetting: it was never accurate in the first place. This illustrates why single-metric (forgetting-only) evaluation is insufficient, and motivates reporting accuracy alongside forgetting.

### 4.3 Phase 11: Mechanism Ablation

To isolate the productive component of TCL, we ran a pre-registered four-condition ablation: (1) full TCL (EMA penalty + governor), (2) penalty-only (EMA penalty, governor disabled), (3) governor-only (LR adjustment, no penalty), and (4) plain SGD as a reference. Results are in Table 3.

**Penalty is the mechanism.** Governor-only performs *worse* than SGD (0.250 vs. 0.219) despite having the same objective, indicating that the regime-detection LR adjustment actively interferes with learning when used without the penalty. Full TCL outperforms governor-only ($\Delta$=−0.123, $p$=0.023, $d$=1.60, 5/5 seeds), a result pre-registered as a falsification test of the hypothesis "governor contributes" and classified as PUBLICATION_ALLOWED in our evidence inventory. This negative finding—the governor mechanism is harmful without the penalty and contributes nothing when combined with it—is the mechanistic conclusion of this paper.

**Governor never activates.** Examination of all production traces confirms $\rho$=0, $\sigma$=0: the gradient-ratio regime detector has never triggered the LR adjustment in any evaluated configuration. This is consistent with the governor-only ablation result.

Table 3: Mechanism ablation (Phase 11, Split-CIFAR-10). $n$=5 seeds. Pre-registered hypothesis: *Does the governor contribute?* Verdict: governor-alone is worse than SGD; penalty is the sole mechanism.

| | Forgetting ↓ | | Accuracy ↑ | |
|---|---|---|---|---|
| Condition | Mean | 95% CI | Mean | 95% CI |
| Full TCL | 0.127 | $[0.100, 0.154]$ | 0.766 | $[0.737, 0.795]$ |
| Penalty-only | 0.143 | $[0.121, 0.165]$ | 0.765 | $[0.748, 0.782]$ |
| Governor-only | 0.250 | $[0.165, 0.335]$ | 0.679 | $[0.606, 0.753]$ |
| SGD | 0.219 | $[0.144, 0.295]$ | 0.709 | $[0.643, 0.775]$ |

Table 4: Pairwise comparisons (Phase 11, full TCL as reference, $n$=5). Pre-registered hypothesis was governor contribution; it is falsified.

| Comparison | $\Delta$ | $p$ (uncorr.) | Cohen's $d$ | Seeds TCL better |
|---|---|---|---|---|
| Full vs. Governor-only | −0.123 | 0.023 | 1.60 | 5/5 |
| Full vs. SGD | −0.092 | 0.020 | 1.68 | 5/5 |
| Full vs. Penalty-only | −0.016 | 0.315 | 0.51 | 4/5 |

Full vs. penalty-only: $p$=0.315, underpowered (24.5% at $d$=0.51, $n$=5). Need $n$=26 for 80% power. Null result is *uninformative*.

**Full vs. penalty-only.** The comparison between full TCL and penalty-only is not statistically resolved ($\Delta$=−0.016, $p$=0.315, $d$=0.51, 4/5 seeds). Power at $n$=5 for $d$=0.51 is only 24.5%, meaning the null result is *uninformative*: we cannot conclude the governor adds zero benefit at $n$=5. Resolving this comparison requires $n$=26 for 80% power. The available evidence supports that the penalty explains *most* of the TCL benefit; the exact magnitude of any residual governor contribution is undetermined.

# 5 Analysis

**EWC default hyperparameters underfit.** EWC with $\lambda$=100 achieves forgetting 0.191, worse than SGD's 0.233 in the direction expected but with a gap that—directionally—favours TCL. The Fisher-EMA theorem suggests the snapshot bias is structural: for a ResNet-18 with 40 epochs per task, early-training gradients (epochs 1–10) are dominated by loss curvature near random initialization, not by the task's genuine parameter importance. Mixing these into a uniform terminal window inflates importance weights for parameters with high early-training sensitivity, over-constraining them in subsequent tasks. The EMA's exponential recency bias corrects this by down-weighting early batches by $\beta^{T-t}$, focusing importance estimation on the convergence phase.

**Effect size comparison.** Cohen's $d$=3.68 for TCL vs. SGD is a very large effect by standard conventions ($d > 0.8$: large; $d > 1.2$: very large). Cohen's $d$=1.70 for TCL vs. EWC is also a large effect. At $n$=5, power for $d$=3.68 is 100% (confirmed headline result); power for $d$=1.70 is 92.5% (adequate, but $p$=0.019 fails the Bonferroni-corrected threshold of 0.0125).

**Family-wise error control.** Phase 10 tests four hypotheses simultaneously (TCL vs. EWC, SI, SGD, and an SI accuracy diagnostic). Without correction, the expected number of false positives at $\alpha$=0.05 is 0.2 per family of four null hypotheses. Bonferroni correction raises the per-comparison threshold to 0.0125. Only the TCL vs. SGD result ($p$=0.0012) passes this threshold. The directional TCL vs. EWC finding motivates the pre-registered EWC replication comparison (Phase 2, not yet run) as a pre-specified single-hypothesis test, which would not require Bonferroni correction.

# 6 Limitations

**1. The thermodynamic governor does not contribute.** The regime-detection LR adjustment component of TCL—the mechanism that originally motivated the "thermodynamic" framing—is empirically falsified as a productive mechanism. Governor-alone produces *higher* forgetting than plain SGD (0.250 vs. 0.219), and full TCL outperforms governor-alone ($p$=0.023, $d$=1.60, 5/5 seeds, pre-registered falsification). The benefit of the full TCL system is attributable to the elastic penalty, not to thermodynamic regime detection.

**2. Regime-detection LR adjustment never fires.** In every production trace examined, the gradient-magnitude ratio $\rho$=0 and the reference magnitude $\sigma$=0. The learning-rate adjustment mechanism has never triggered in any evaluated configuration. This means all reported TCL results were produced without the LR adjustment, and the reported performance reflects the elastic penalty alone.

**3. Task-incremental protocol only.** All experiments provide task identity at test time (task-incremental scenario). Results do not extend to class-incremental evaluation (where task identity is withheld) without separate experimentation. Class-incremental evaluation is substantially harder, and methods that reduce forgetting in task-incremental settings often fail to generalize.

**4. Single backbone (ResNet-18).** All results are obtained with ResNet-18. Generalization to Vision Transformers, wider residual networks, or other architectures is untested. The EMA importance accumulator is architecture-agnostic in principle, but empirical validation requires separate evaluation.

**5. TCL vs. EWC does not survive Bonferroni correction.** The TCL vs. EWC comparison ($p$=0.019, $d$=1.70, 5/5 seeds) does not survive family-wise correction at $k$=4 (Bonferroni threshold $p$=0.0125). This result is directional evidence only. It motivates further pre-registered replication as a single-hypothesis test (Phase 2, pre-registered but not yet run), which would not require Bonferroni correction and would provide definitive confirmation or disconfirmation.

**6. DER++ and LwF baselines not yet included.** Dark Experience Replay (DER++) and Learning without Forgetting (LwF) are not included in the current comparison. Phase 2 GPU experiments (pre-registered, scripts written) include both baselines with fair hyperparameter selection, but those experiments have not yet been executed. Current results should therefore not be interpreted as establishing TCL's position in a complete benchmark; they establish the EMA mechanism finding and the SGD gap specifically.

# 7 Conclusion

We have presented a formal and empirical argument that EWC's terminal-snapshot importance estimator is suboptimal under non-stationary training gradients, and that replacing it with a continuous exponential moving average—the ThermalImportance accumulator—corrects this bias. The resulting method, TCL, reduces catastrophic forgetting on Split-CIFAR-10 by 0.117 relative to SGD ($p$=0.0012, $d$=3.68, Bonferroni-corrected), and shows a large directional improvement over EWC with default hyperparameters ($\Delta$=$-$0.075, $d$=1.70, 5/5 seeds, directional pending replication).

A pre-registered mechanism ablation definitively falsifies the hypothesis that TCL's thermodynamic governor component contributes to performance. Governor-alone performs worse than SGD; the elastic penalty is the entire operative mechanism.

The honest scope of these results is narrow: task-incremental only, ResNet-18 only, Split-CIFAR-10 only. The confirmed headline finding is the SGD gap. The EWC gap requires pre-registered replication (Phase 2 scripts written and ready). We report this work at workshop stage with the intention of completing the replication before any archival submission.

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